Homework 2

Branda Sisoutham

2026-09-03

Question 1 (C2 E2)

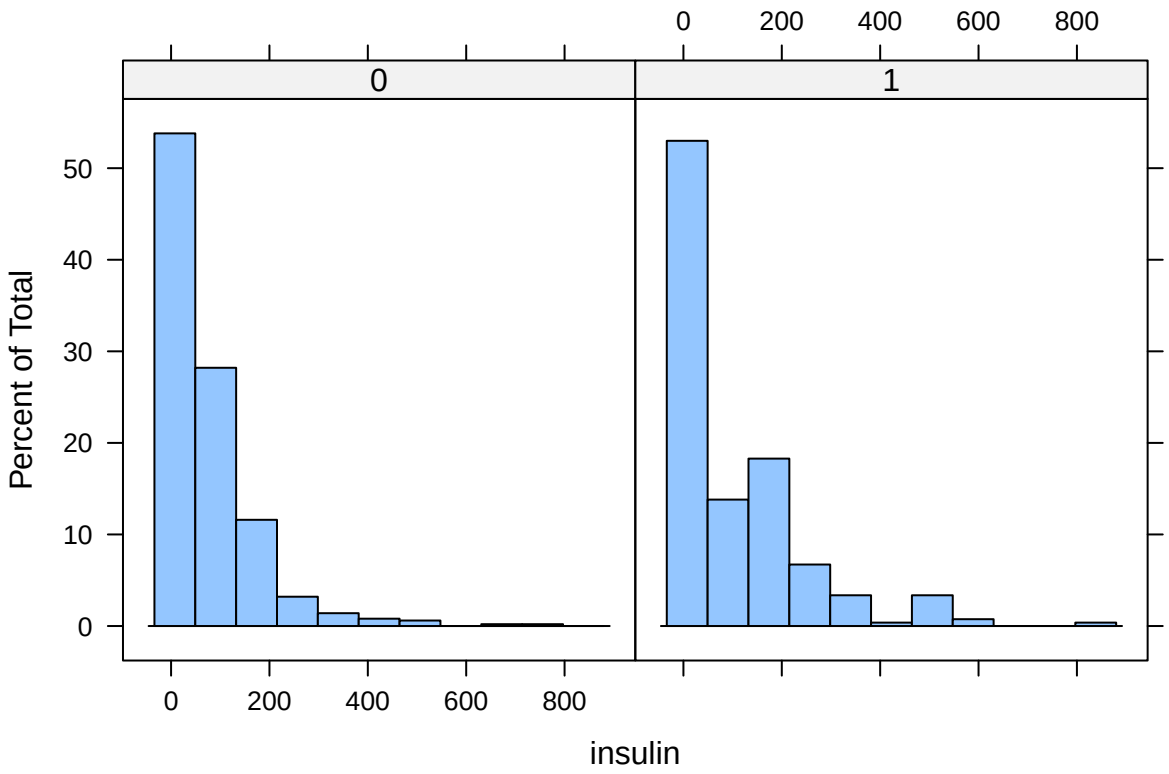
```
head(pima)
```

```
##   pregnant glucose diastolic triceps insulin  bmi diabetes age test
## 1        6     148        72      35       0 33.6    0.627  50    1
## 2        1      85        66      29       0 26.6    0.351  31    0
## 3        8     183        64       0       0 23.3    0.672  32    1
## 4        1      89        66      23      94 28.1    0.167  21    0
## 5        0     137        40      35     168 43.1    2.288  33    1
## 6        5     116        74       0       0 25.6    0.201  30    0
```

```
pima$testf = factor(pima$test)
summary(pima$testf)
```

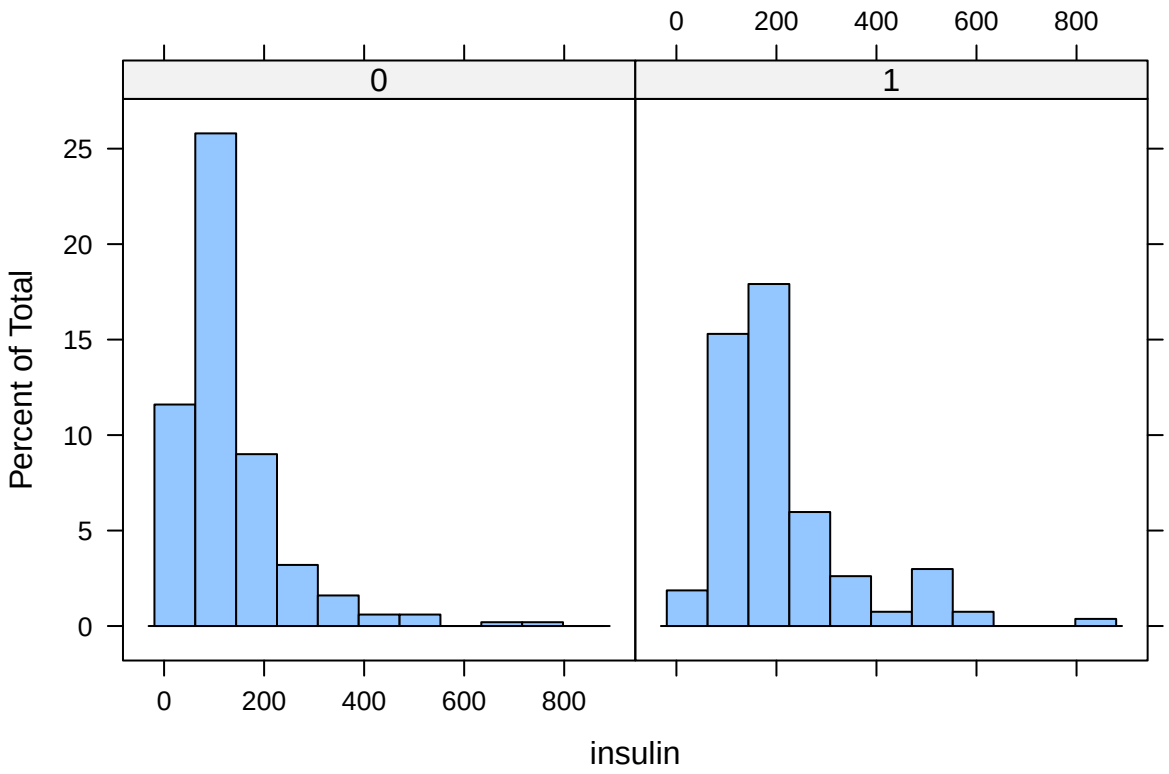
```
##   0   1
## 500 268
```

```
histogram(~ insulin | testf, data = pima)
```



The distributions are right skewed for both groups, much of the data being 0 which could mean some values are missing.

```
pima$insulin[pima$insulin == 0] = NA
histogram(~ insulin | testf, data = pima)
```



Both the distributions are still right skewed, this means women who tested positive for diabetes tend to have higher insulin levels than those who tested negative.

```
pima$glucose[pima$glucose == 0] = NA
pima$diastolic[pima$diastolic == 0] = NA
pima$triceps[pima$triceps == 0] = NA
pima$bmi[pima$bmi == 0] = NA
m1 = lm(test ~ pregnant + glucose + diastolic + triceps +
        insulin + bmi + diabetes + age, data=pima)

summary(m1)
```

```
##
## Call:
## lm(formula = test ~ pregnant + glucose + diastolic + triceps +
##     insulin + bmi + diabetes + age, data = pima)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.07966 -0.25711 -0.06177  0.25851  1.03750
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.103e+00  1.436e-01  -7.681 1.34e-13 ***
## pregnant     1.295e-02  8.364e-03   1.549  0.12230
## glucose      6.409e-03  8.159e-04   7.855 4.07e-14 ***
```

```
## diastolic    5.465e-05  1.730e-03   0.032  0.97482
## triceps      1.678e-03  2.522e-03   0.665  0.50631
## insulin     -1.233e-04  2.045e-04  -0.603  0.54681
## bmi          9.325e-03  3.901e-03   2.391  0.01730 *
## diabetes     1.572e-01  5.804e-02   2.708  0.00707 **
## age          5.878e-03  2.787e-03   2.109  0.03559 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3853 on 383 degrees of freedom
##   (376 observations deleted due to missingness)
## Multiple R-squared:  0.3458, Adjusted R-squared:  0.3321
## F-statistic:  25.3 on 8 and 383 DF,  p-value: < 2.2e-16
```

The model uses 392 observations which is less because the model does not use data that has missing variables.

```
m2 = lm(test ~ pregnant + glucose + diastolic + triceps +
        bmi + diabetes + age, data=pima)

summary(m2)
```

```
##
## Call:
## lm(formula = test ~ pregnant + glucose + diastolic + triceps +
##     bmi + diabetes + age, data = pima)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.17935 -0.26403 -0.06998  0.27128  1.00835
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.0372591  0.1189123  -8.723  < 2e-16 ***
## pregnant     0.0201144  0.0065716   3.061 0.002321 **
## glucose      0.0060256  0.0005831  10.334  < 2e-16 ***
## diastolic   -0.0009059  0.0015195  -0.596 0.551318
## triceps      0.0006001  0.0021024   0.285 0.775433
## bmi          0.0118797  0.0033199   3.578 0.000378 ***
## diabetes     0.1803291  0.0494867   3.644 0.000295 ***
## age          0.0042948  0.0021671   1.982 0.048021 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3834 on 524 degrees of freedom
##   (236 observations deleted due to missingness)
## Multiple R-squared:  0.348,  Adjusted R-squared:  0.3392
## F-statistic: 39.95 on 7 and 524 DF,  p-value: < 2.2e-16
```

When we remove insulin, the model used 532 observation and its more than the old model because even from the histogram we can see that many of them has missing insulin values.

```r
pima_complete = model.frame(m1)
m2_same = lm(test ~ pregnant + glucose + diastolic + triceps +
             bmi + diabetes + age, data=pima_complete)
anova(m2_same, m1)
```

```
## Analysis of Variance Table
##
## Model 1: test ~ pregnant + glucose + diastolic + triceps + bmi + diabetes +
##     age
## Model 2: test ~ pregnant + glucose + diastolic + triceps + insulin + bmi +
##     diabetes + age
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    384 56.898
## 2    383 56.844  1  0.053982 0.3637 0.5468
```

Using an F-test for the models using the same observations since they had different amounts of them, there is not enough evidence that insulin significantly improves the model since out p-value is $0.54 > 0.05$.

```r
m_full = lm(test ~ pregnant + glucose + diastolic + triceps +
            insulin + bmi + diabetes + age,
            data = pima_complete)
m_aic = step(m_full, direction = "backward")
```

```
## Start:  AIC=-738.93
## test ~ pregnant + glucose + diastolic + triceps + insulin + bmi +
##     diabetes + age
##
##             Df Sum of Sq    RSS     AIC
## - diastolic  1    0.0001 56.844 -740.93
## - insulin    1    0.0540 56.898 -740.56
## - triceps    1    0.0657 56.910 -740.48
## <none>                   56.844 -738.93
## - pregnant   1    0.3559 57.200 -738.48
## - age        1    0.6602 57.504 -736.40
## - bmi        1    0.8483 57.693 -735.12
## - diabetes   1    1.0886 57.933 -733.49
## - glucose    1    9.1576 66.002 -682.38
##
## Step:  AIC=-740.93
## test ~ pregnant + glucose + triceps + insulin + bmi + diabetes +
##     age
##
##             Df Sum of Sq    RSS     AIC
## - insulin    1    0.0546 56.899 -742.55
## - triceps    1    0.0657 56.910 -742.48
## <none>                   56.844 -740.93
## - pregnant   1    0.3570 57.201 -740.48
## - age        1    0.6850 57.529 -738.23
## - bmi        1    0.9042 57.749 -736.74
## - diabetes   1    1.0962 57.941 -735.44
## - glucose    1    9.2587 66.103 -683.78
##
```

```
## Step:  AIC=-742.55
## test ~ pregnant + glucose + triceps + bmi + diabetes + age
##
##            Df Sum of Sq    RSS     AIC
## - triceps   1    0.0663 56.965 -744.10
## <none>                  56.899 -742.55
## - pregnant  1    0.3777 57.277 -741.96
## - age       1    0.6640 57.563 -740.00
## - bmi       1    0.8706 57.770 -738.60
## - diabetes  1    1.0758 57.975 -737.21
## - glucose   1   11.8228 68.722 -670.55
##
## Step:  AIC=-744.1
## test ~ pregnant + glucose + bmi + diabetes + age
##
##            Df Sum of Sq    RSS     AIC
## <none>                  56.965 -744.10
## - pregnant  1    0.3963 57.362 -743.38
## - age       1    0.6974 57.663 -741.33
## - diabetes  1    1.1147 58.080 -738.50
## - bmi       1    2.0904 59.056 -731.97
## - glucose   1   11.8737 68.839 -671.88
```

Using backward selection with AIC, the selected predictors were pregnant, glucose, BMI, diabetes, and age. The selected model used 392 observations.

```
pima$missing = as.numeric(!complete.cases(pima))
table(pima$missing)
```

```
##
##   0   1
## 392 376
```

```
m_missing = lm(test ~ missing, data=pima)
summary(m_missing)
```

```
##
## Call:
## lm(formula = test ~ missing, data = pima)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.3670 -0.3670 -0.3316  0.6330  0.6684
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.33163    0.02409  13.767   <2e-16 ***
## missing      0.03539    0.03443   1.028    0.304
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4769 on 766 degrees of freedom
## Multiple R-squared:  0.001378,   Adjusted R-squared:  7.383e-05
## F-statistic: 1.057 on 1 and 766 DF,  p-value: 0.3043
```

```r
m_final = lm(test ~ pregnant + glucose + bmi + diabetes + age,
             data=pima)

summary(m_final)
```

```
##
## Call:
## lm(formula = test ~ pregnant + glucose + bmi + diabetes + age,
##     data = pima)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1415 -0.2918 -0.0750  0.3010  1.0085
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.0640595  0.0869491 -12.238  < 2e-16 ***
## pregnant     0.0194424  0.0051109   3.804 0.000154 ***
## glucose      0.0063476  0.0005053  12.562  < 2e-16 ***
## bmi          0.0135780  0.0021618   6.281 5.71e-10 ***
## diabetes     0.1387755  0.0445520   3.115 0.001910 **
## age          0.0017978  0.0015138   1.188 0.235360
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3956 on 746 degrees of freedom
##   (16 observations deleted due to missingness)
## Multiple R-squared:  0.3187, Adjusted R-squared:  0.3141
## F-statistic: 69.79 on 5 and 746 DF,  p-value: < 2.2e-16
```

Using the test, the missing values was not significantly associated with diabetes test results based on our p-value. Based off that, I refit the selected model using the full dataset, which is 752 observations instead of 392. This is appropriate because the selected model does not include the variables responsible for most of the missing values.

```r
quantile(pima$bmi, c(0.25, 0.75), na.rm=TRUE)
```

```
##  25%  75%
## 27.5 36.6
```

```r
coef(m_final)["bmi"] * (36.6 - 27.5)
```

```
##       bmi
## 0.1235603
```

```r
confint(m_final, "bmi") * (36.6 - 27.5)
```

```
##          2.5 %    97.5 %
## bmi 0.08494094 0.1621796
```

A woman at the third BMI quartile has an estimated probability of testing positive for diabetes which is 12.36 percentage points higher than a woman in the first quartile. The 95% confidence interval for this difference is 8.49 to 16.22 percentage points.

```
t.test(diastolic ~ testf, data=pima)
```

```
##
##  Welch Two Sample t-test
##
## data:  diastolic by testf
## t = -4.6643, df = 504.72, p-value = 3.972e-06
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -6.316023 -2.572156
## sample estimates:
## mean in group 0 mean in group 1
##        70.87734        75.32143
```

Women who tested positive had a higher mean diastolic blood pressure of 75.32 than woman who tested negative 70.88, and the t-test showed a significant difference between p-value. However, diastolic blood pressure was no significant in the full regression model. This is not contradictory because the t-test looks at diastolic blood pressure by itself, while regression measures its effect after account the other predictors.

Question 2 (C2 E3)

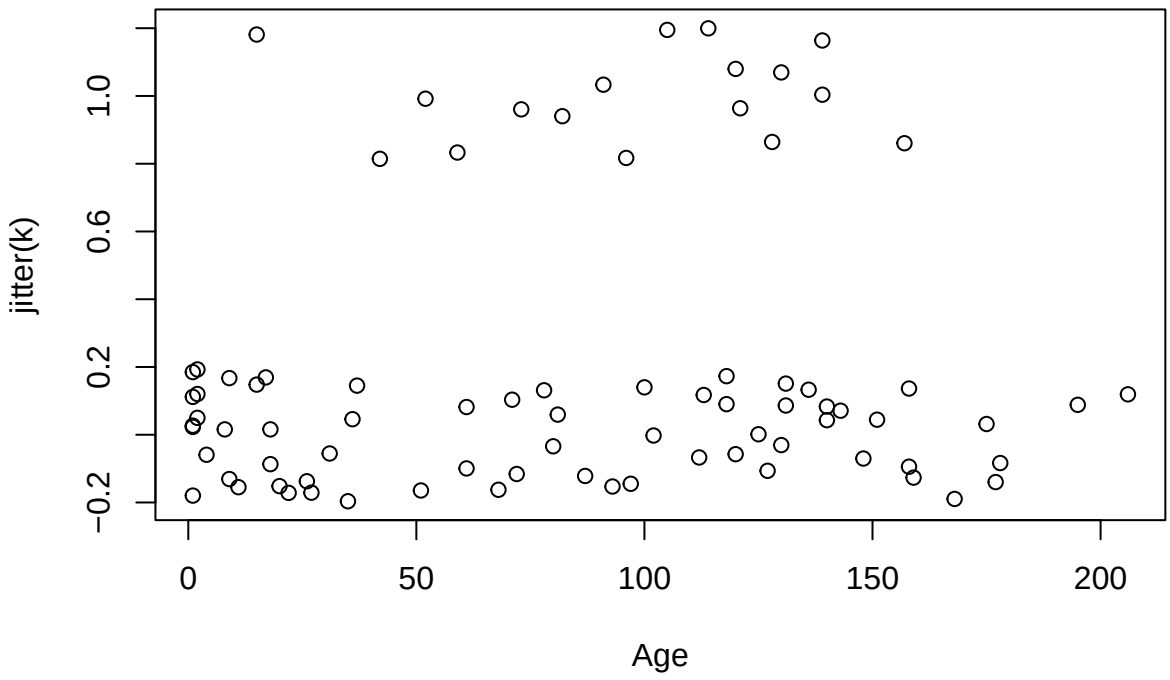
```
data(kyphosis, package="rpart")
```

```
data(kyphosis, package="rpart")
head(kyphosis)
```

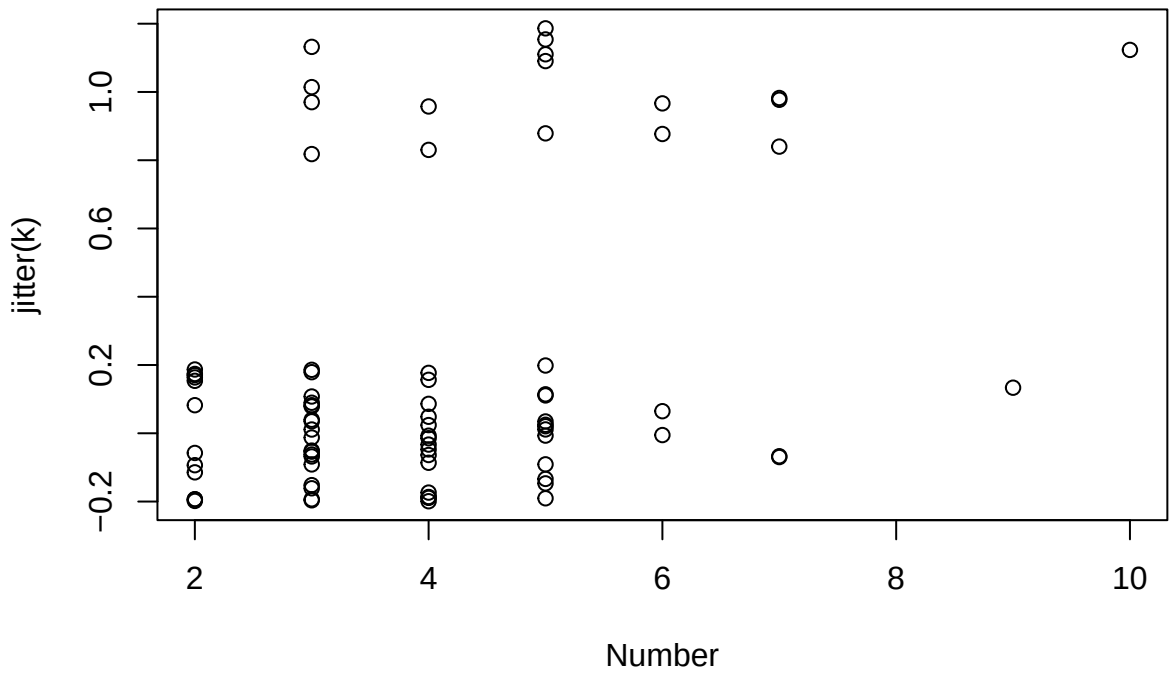
```
##   Kyphosis Age Number Start
## 1   absent  71      3     5
## 2   absent 158      3    14
## 3  present 128      4     5
## 4   absent   2      5     1
## 5   absent   1      4    15
## 6   absent   1      2    16
```

```
k = as.numeric(kyphosis$Kyphosis == "present")
```

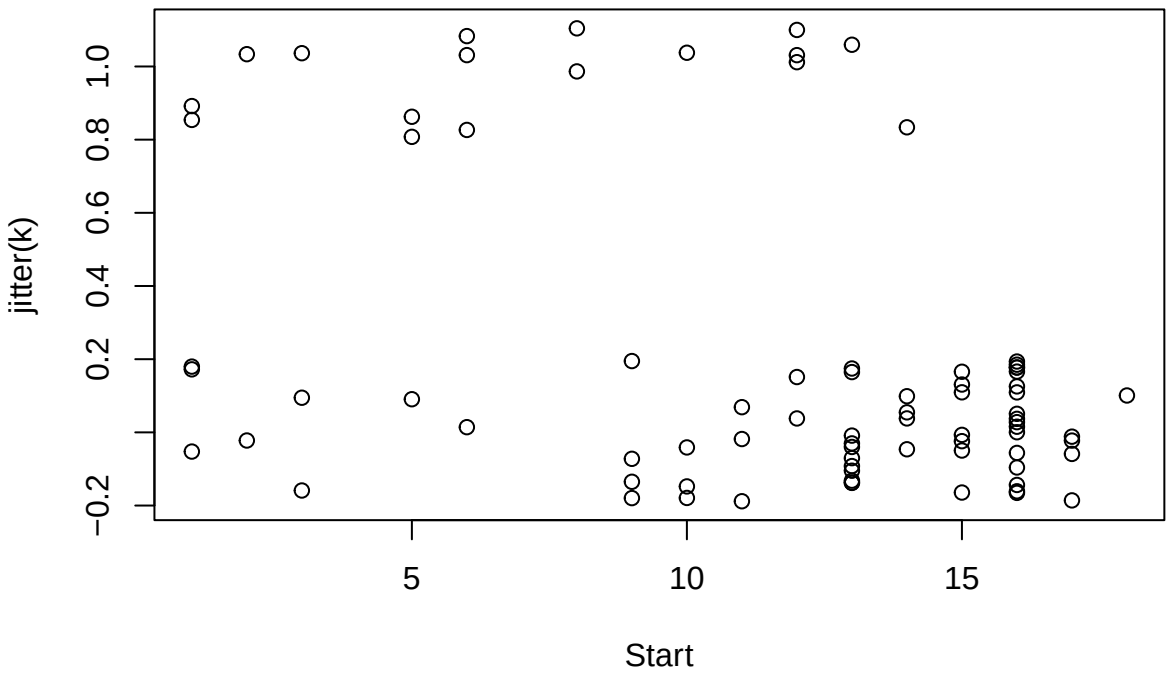
```
plot(jitter(k) ~ Age, data=kyphosis)
```




```
plot(jitter(k) ~ Number, data=kyphosis)
```




```
plot(jitter(k) ~ Start, data=kyphosis)
```



All three predictors have some relationship with kyphosis and it appears to be more common with higher ageand Number vales and lower Start values.

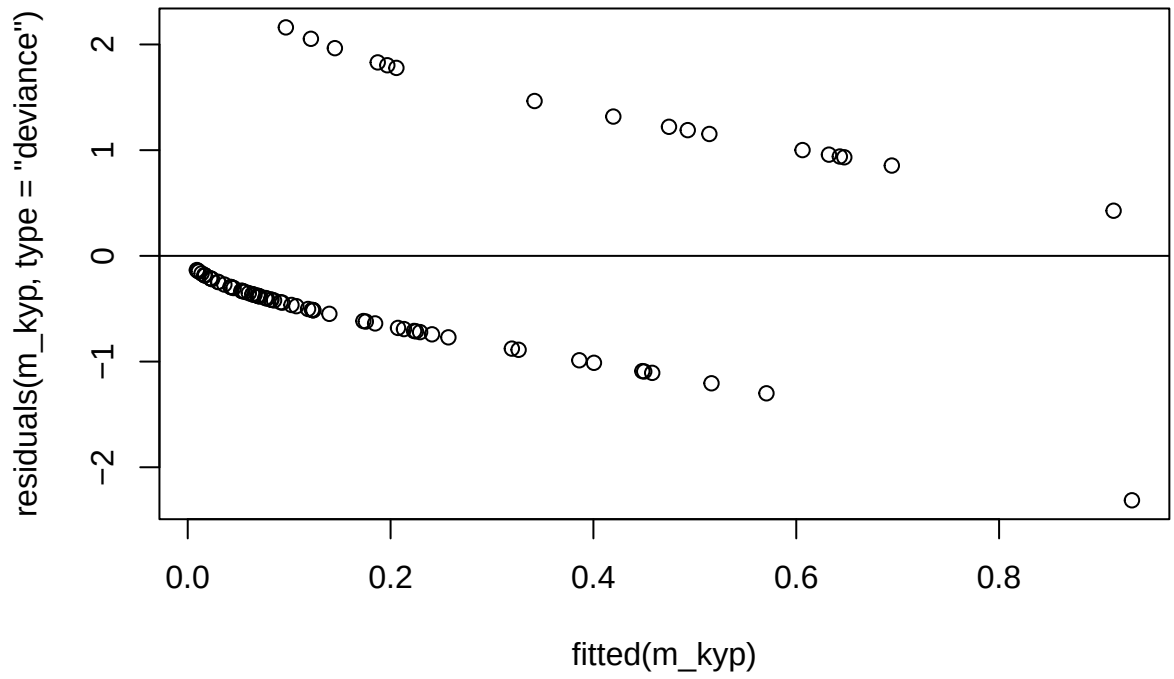
```
m_kyp = glm(Kyphosis ~ Age + Number + Start,
            family = binomial,
            data = kyphosis)

summary(m_kyp)
```

```
##
## Call:
## glm(formula = Kyphosis ~ Age + Number + Start, family = binomial,
##     data = kyphosis)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.036934   1.449575  -1.405  0.15996
## Age          0.010930   0.006446   1.696  0.08996 .
## Number       0.410601   0.224861   1.826  0.06785 .
## Start       -0.206510   0.067699  -3.050  0.00229 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 83.234  on 80  degrees of freedom
```

```
## Residual deviance: 61.380  on 77  degrees of freedom
## AIC: 69.38
##
## Number of Fisher Scoring iterations: 5
```

```
plot(fitted(m_kyp), residuals(m_kyp, type="deviance"))
abline(h=0)
```

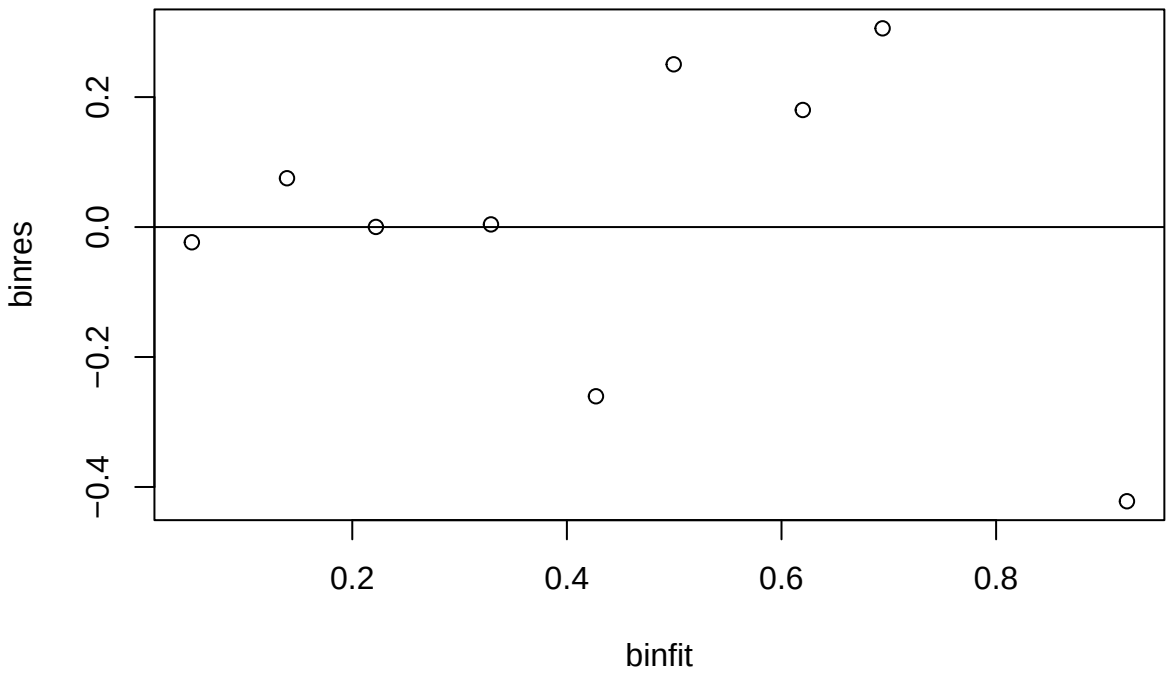


The residual plot shows two distinct groups because the response is binary. Since a few of the observations have large residuals this could mean that the predctions may be a little off.

```
bins = cut(fitted(m_kyp), breaks=10)

binfit = tapply(fitted(m_kyp), bins, mean)
binres = tapply(residuals(m_kyp, type="response"), bins, mean)

plot(binfit, binres)
abline(h=0)
```

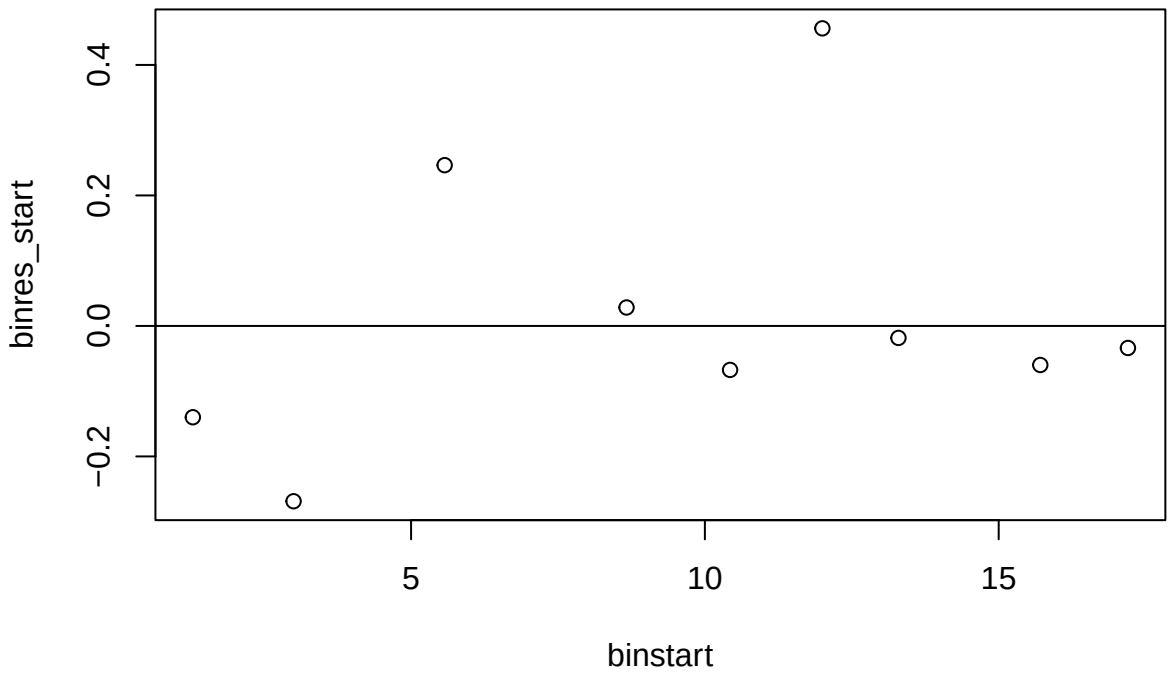


Many of the residuals hover around 0 but the others are near 0.2 especially the higher fitted values. The model may not fit well for the observations.

```
bins_start = cut(kyphosis$Start, breaks=10)

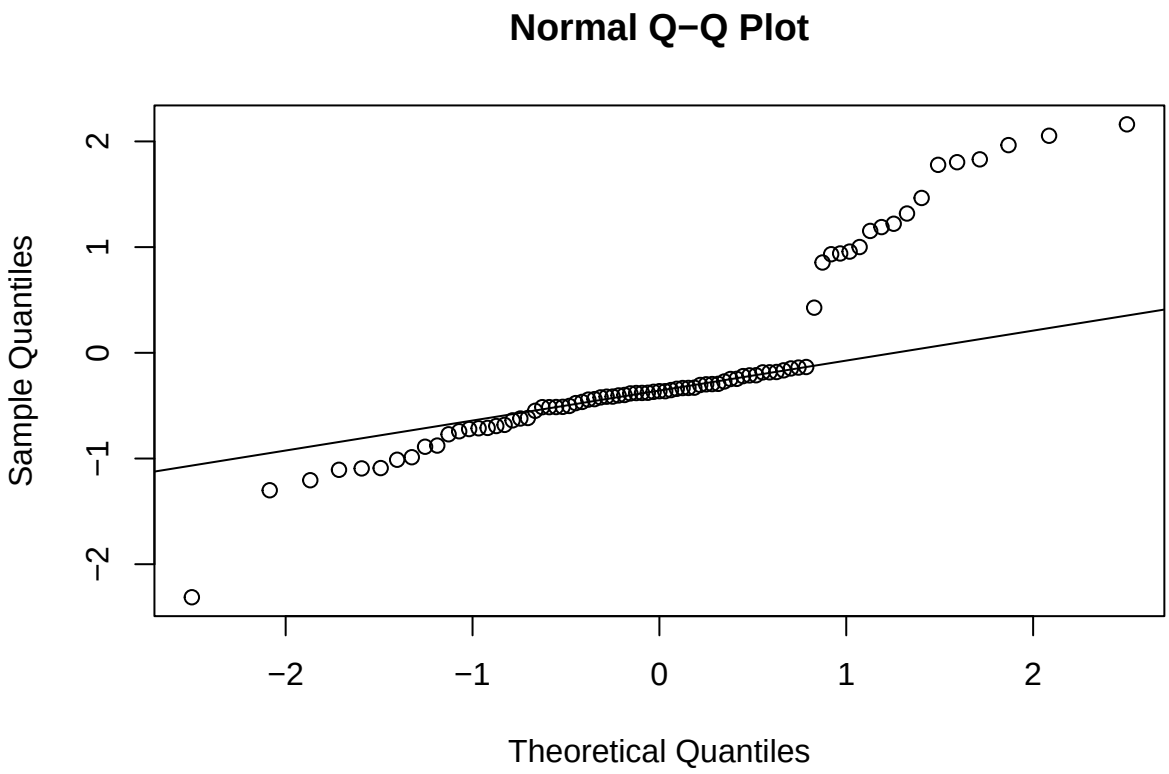
binstart = tapply(kyphosis$Start, bins_start, mean)
binres_start = tapply(residuals(m_kyp, type="response"), bins_start, mean)

plot(binstart, binres_start)
abline(h=0)
```



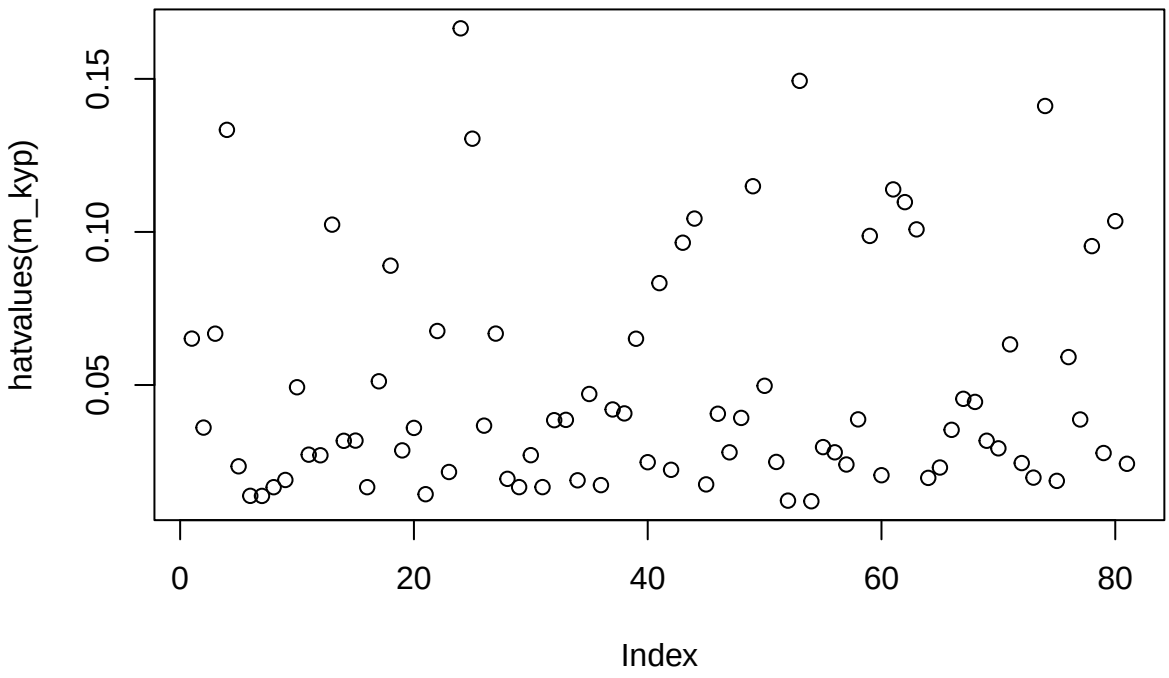
Since the higher fitted values are starting to hover around 0 this could mean that the response may not be linear.

```
qqnorm(residuals(m_kyp, type="deviance"))
qqline(residuals(m_kyp, type="deviance"))
```



The residuals dont follow a straight line so this means that they are not normally distributed.

```
plot(hatvalues(m_kyp))
```



Most of the observations have low leverage, there are a couple have have high leverage and these could be the ones influencing the fitted model.

```
library(ResourceSelection)
```

```
## Warning: package 'ResourceSelection' was built under R version 4.4.3
```

```
## ResourceSelection 0.3-6   2023-06-27
```

```
hoslem.test(as.numeric(kyphosis$Kyphosis == "present"),
            fitted(m_kyp),
            g = 10)
```

```
##
##  Hosmer and Lemeshow goodness of fit (GOF) test
##
## data:  as.numeric(kyphosis$Kyphosis == "present"), fitted(m_kyp)
## X-squared = 6.3464, df = 8, p-value = 0.6085
```

```
pred = fitted(m_kyp)
obs = as.numeric(kyphosis$Kyphosis == "present")

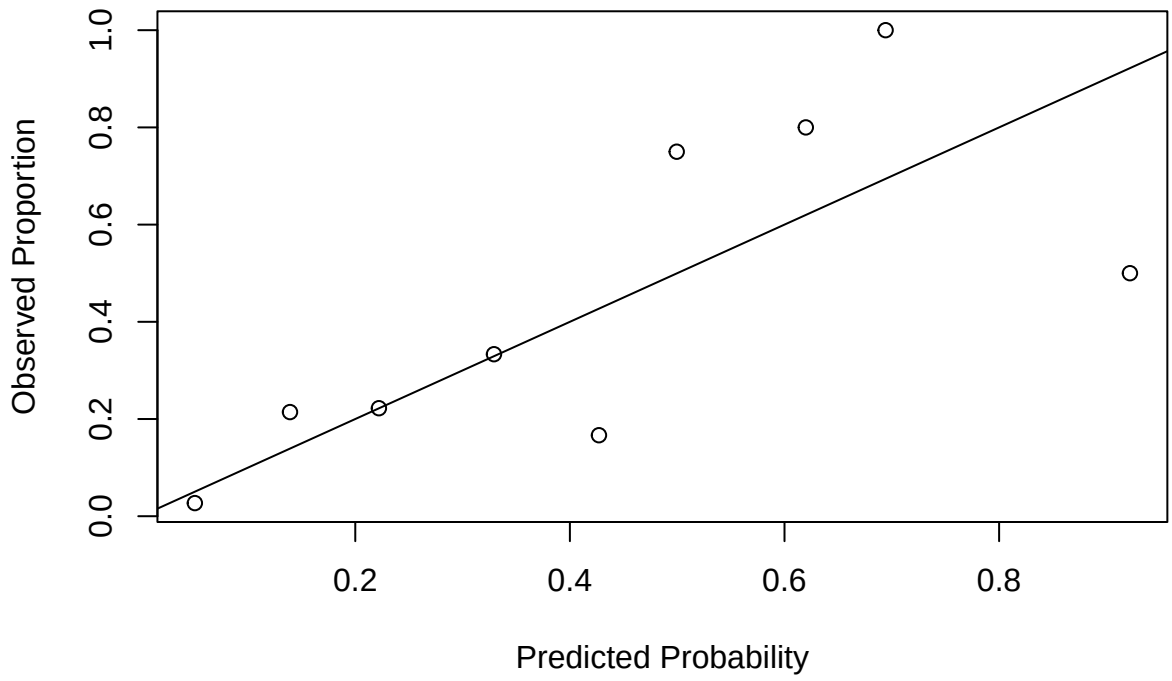
groups = cut(pred, breaks=10)

pred_bin = tapply(pred, groups, mean)
```

```r
obs_bin = tapply(obs, groups, mean)

plot(pred_bin, obs_bin,
     xlab="Predicted Probability",
     ylab="Observed Proportion")

abline(0, 1)
```



The observed proportion are somewhat close to the predicted probabilities, since the p value of the Hosmer-Lemeshow was 0.60 theres no strong evidence of the poor model fit.

```r
predicted = ifelse(fitted(m_kyp) > 0.5, "present", "absent")
table(Actual = kyphosis$Kyphosis, Predicted = predicted)
```

```
##          Predicted
## Actual    absent present
##   absent      61       3
##   present     10       7
```

The model correctly predicts present only 41% of the time.